

## VIVAFLOC CP610

Revision: 02

Revision date 19/02/2022

| SECTION 1: Identification of the substance/mixture and of the company/undertaking |                                                                               |                                                                                                                                                               |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1                                                                               | Product identifier                                                            | VIVAFLOC CP610                                                                                                                                                |
| 1.2                                                                               | Relevant identified uses of the substance or mixture and uses advised against |                                                                                                                                                               |
|                                                                                   | Relevant identified uses:                                                     | Retention and drainage aid                                                                                                                                    |
| 1.3                                                                               | Details of the supplier of the safety data sheet                              |                                                                                                                                                               |
|                                                                                   | Company:                                                                      | VIV INTERCHEM<br>22 Sukhumvit 42, Sukhumvit Rd, Prakanong, Klongtoey, Bangkok 10110<br>Telephone: +662 712 1044<br>E-mail address: thasanaprapha@vivgroup.com |
| 1.4                                                                               | Emergency telephone number                                                    |                                                                                                                                                               |
|                                                                                   | International emergency number:                                               | Telephone: 02-7094600-4 #214                                                                                                                                  |

| SECTION 2: Hazards Identification |                                                                                                                                                                                                                                            |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.1                               | Classification of the substance or mixture                                                                                                                                                                                                 |
|                                   | According to Regulation (EC) No 1272/2008 [CLP]<br>No need for classification according to GHS criteria for this product.                                                                                                                  |
| 2.2                               | Label elements                                                                                                                                                                                                                             |
|                                   | According to Regulation (EC) No 1272/2008 [CLP]<br>Hazard pictogram(s): None<br>Signal word: None<br>Hazard statement(s): None<br>Precautionary statement(s): None<br>Additional elements: EUH210 – Safety data sheet available on request |
| 2.3                               | Other hazards                                                                                                                                                                                                                              |
|                                   | According to Regulation (EC) No 1272/2008 [CLP]<br>Very slippery when wet.<br>PBT and vPvB assessment: Not PBT or vPvB according to the criteria of Annex XIII of REACH.                                                                   |

| SECTION 3: Composition/information on ingredients |                                                                                                 |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 3.1                                               | Substances                                                                                      |
|                                                   | Not applicable, this product is a mixture.                                                      |
| 3.2                                               | Mixtures                                                                                        |
|                                                   | Hazardous components                                                                            |
|                                                   | Ethanaminium, N,N,N-trimethyl-2-[(oxo-2-propen-1-yl)oxy]-, chloride, polymer with 2-propenamide |

**VIVAFLOC CP610**

Revision: 02

Revision date 19/02/2022

|  |  |                                |                                           |
|--|--|--------------------------------|-------------------------------------------|
|  |  | Concentration                  | range: >80%                               |
|  |  | CAS Number                     | 69418-26-4                                |
|  |  | Classification according to    | Aquatic Acute 3; H402                     |
|  |  | Chinese Standard GB 30000-2013 |                                           |
|  |  | <b>Adipic acid</b>             |                                           |
|  |  | Concentration                  | 2.5 – 10%                                 |
|  |  | CAS Number                     | 124-04-9                                  |
|  |  | Classification according to    | Eye Irrit. 2A:H319, Aquatic Acute 3; H402 |
|  |  | Chinese Standard GB 30000-2013 |                                           |

**SECTION 4: First aid measures**

|            |                                                                                   |                                                                                                                              |
|------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>4.1</b> | <b>Description of first aid measures</b>                                          |                                                                                                                              |
|            | If inhaled:                                                                       | Move to fresh air. Get medical attention if symptoms occur.                                                                  |
|            | On skin contact:                                                                  | Wash off with soap and plenty of water. Get medical attention if irritation develops and persists.                           |
|            | On contact with eyes:                                                             | Rinse immediately with plenty of water, also under the eyelids. Get medical attention.                                       |
|            | On ingestion:                                                                     | Rinse mouth. If conscious, give the victim plenty of water to drink. Induce vomiting, but only if victim is fully conscious. |
| <b>4.2</b> | <b>Most important symptoms and effects, both acute and delayed</b>                |                                                                                                                              |
|            | Contact with dust can cause mechanical irritation or drying of the skin.          |                                                                                                                              |
| <b>4.3</b> | <b>Indication of any immediate medical attention and special treatment needed</b> |                                                                                                                              |
|            | Other information:                                                                | No information available.                                                                                                    |

**SECTION 5: Firefighting measures**

|            |                                                              |                                                                                                                                                                                                                |
|------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>5.1</b> | <b>Extinguishing media</b>                                   |                                                                                                                                                                                                                |
|            | Suitable extinguishing media:                                | Water. Water spray. Foam. Carbon dioxide (CO <sub>2</sub> ). Dry powder. Warning! Aqueous solutions or powders that become wet render surfaces extremely slippery.                                             |
|            | Unsuitable extinguishing media for safety reasons:           | None                                                                                                                                                                                                           |
| <b>5.2</b> | <b>Special hazards arising from the substance or mixture</b> |                                                                                                                                                                                                                |
|            | Hazardous decomposition products:                            | Hydrogen chloride gas, nitrogen oxides (NO <sub>x</sub> ), carbon oxides (CO <sub>x</sub> ). Hydrogen cyanide (Hydrocyanic acid) may be produced in the event of combustion in an Oxygen deficient atmosphere. |

## VIVAFLOC CP610

Revision: 02

Revision date 19/02/2022

|            |                                    |                                                                                  |
|------------|------------------------------------|----------------------------------------------------------------------------------|
|            | Thermal decomposition may produce: |                                                                                  |
| <b>5.3</b> | <b>Advice for fire-fighters</b>    |                                                                                  |
|            | Protective measures:               | Wear a self-contained breathing apparatus for fire fighting if necessary.        |
|            | Other information:                 | Aqueous solutions or powders that become wet render surfaces extremely slippery. |

### SECTION 6: Accidental release measures

|            |                                                                                                                           |                                                                                                                  |
|------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>6.1</b> | <b>Personal precautions, protective equipment and emergency procedures</b>                                                |                                                                                                                  |
|            | Personal precautions:                                                                                                     | Aqueous solutions or powders that become wet render surfaces extremely slippery.                                 |
|            | Protective equipment:                                                                                                     | Wear adequate personal protective equipment.                                                                     |
|            | Emergency procedures:                                                                                                     | Keep people away from spill/leak. Prevent further leakage or spillage if safe to do so.                          |
| <b>6.2</b> | <b>Environmental precautions</b>                                                                                          | Do not discharge into drains/surface waters/groundwater.                                                         |
| <b>6.3</b> | <b>Methods and material for containment and cleaning up</b>                                                               |                                                                                                                  |
|            | Small spills:                                                                                                             | Do not flush with water. Clean up promptly by sweeping or vacuum.                                                |
|            | Large spills:                                                                                                             | Do not flush with water. Prevent unauthorized access. Sweep up and shovel into suitable containers for disposal. |
|            | Residues:                                                                                                                 | Sweep up to prevent slip hazard. After cleaning, flush away traces with water.                                   |
| <b>6.4</b> | <b>Reference to other sections</b>                                                                                        |                                                                                                                  |
|            | Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13. |                                                                                                                  |

### SECTION 7: Handling and storage

|            |                                                                     |                                                                                                                                                                                                                    |
|------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>7.1</b> | <b>Precautions for safe handling</b>                                | Aqueous solutions or powders that become wet render surfaces extremely slippery. Avoid contact with skin and eyes. Avoid dust formation. Avoid breathing dust. Wash hands before breaks and at the end of workday. |
| <b>7.2</b> | <b>Conditions for safe storage, including any incompatibilities</b> |                                                                                                                                                                                                                    |
|            | Keep in a dry place. Incompatible with oxidizing agents.            |                                                                                                                                                                                                                    |
|            | Storage stability:                                                  | Avoid extreme heat.                                                                                                                                                                                                |
| <b>7.3</b> | <b>Specific end use(s)</b>                                          | For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.                                                                                                   |

### SECTION 8: Exposure controls/personal protection

|            |                                                      |                                              |
|------------|------------------------------------------------------|----------------------------------------------|
| <b>8.1</b> | <b>Components with occupational exposure limits:</b> |                                              |
|            | <b>Control parameters</b>                            | Components with occupational exposure limits |

## VIVAFLOC CP610

Revision: 02

Revision date 19/02/2022

|            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                                        | <p>Particles, not otherwise specified, respirable</p> <p>Particles, not otherwise specified, inhalable</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>8.2</b> | <b>Personal Protective Equipment :</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|            | Eye/Face Protection:                   | Safety glasses with side-shields (frame goggles) (e.g. EN 166)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|            | Body Protection:                       | light protective clothing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|            | Hand protection:                       | <p>Chemical resistant protective gloves (EN 374) Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding &gt; 480 minutes of permeation time according to EN 374): e.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), polyvinylchloride (0.7 mm) and other Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.</p> |
|            | Respiratory Protection:                | <p>Suitable respiratory protection for lower concentrations or short-term effect: Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|            | General safety and hygiene measures:   | Handle in accordance with good industrial hygiene and safety practice. Ensure adequate ventilation. Wearing of closed work clothing is recommended. No eating, drinking, smoking or tobacco use at the place of work.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|            | Environmental exposure controls:       | For information regarding environmental exposure controls, see Section 6.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## SECTION 9: Physical and chemical properties

|            |                                                              |                                      |
|------------|--------------------------------------------------------------|--------------------------------------|
| <b>9.1</b> | <b>Information on basic physical and chemical properties</b> |                                      |
|            | Appearance:                                                  | Granular solid                       |
|            | Colour:                                                      | White                                |
|            | Odour:                                                       | None                                 |
|            | Odour threshold:                                             | No applicable information available. |
|            | pH value:                                                    | 2.5 - 4.5 (5 g/l)                    |
|            | Melting range:                                               | >100 °C                              |
|            | Boiling point:                                               | Not applicable                       |
|            | Flash point:                                                 | Not applicable                       |
|            | Evaporation rate:                                            | The product is a non-volatile solid. |
|            | Flammability (solid/gas):                                    | Not highly flammable                 |

## VIVAFLOC CP610

Revision: 02

Revision date 19/02/2022

|            |                                                     |                                                                                                   |
|------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------|
|            | Lower explosion limit:                              | For solids not relevant for classification and labelling.                                         |
|            | Upper explosion limit:                              | For solids not relevant for classification and labelling.                                         |
|            | Vapour pressure:                                    | The product has not been tested                                                                   |
|            | Solubility in water:                                | Soluble in water                                                                                  |
|            | Partitioning coefficient n-octanol/water (log Kow): | Study scientifically not justified.                                                               |
|            | Self ignition:                                      | Not self-igniting                                                                                 |
|            | Decomposition temperature:                          | > 200°                                                                                            |
|            | Viscosity, dynamic:                                 | Not determined                                                                                    |
|            | Explosion hazard:                                   | Not expected to be explosive based on the chemical structure.                                     |
|            | Oxidizing properties:                               | Not expected to be explosive based on the chemical structure.                                     |
| <b>9.2</b> | <b>Other information</b>                            |                                                                                                   |
|            | Self heating ability:                               | It is not a substance capable of spontaneous heating.                                             |
|            | Bulk density:                                       | Approx. 900 kg/m <sup>3</sup>                                                                     |
|            | Hygroscopy:                                         | Non-hygroscopic                                                                                   |
|            | Other Information:                                  | If necessary, information on other physical and chemical parameters is indicated in this section. |

### SECTION 10: Stability and reactivity

|             |                                           |                                                                          |
|-------------|-------------------------------------------|--------------------------------------------------------------------------|
| <b>10.1</b> | <b>Reactivity</b>                         | Hazardous polymerization does not occur.                                 |
| <b>10.2</b> | <b>Chemical stability</b>                 | Stable                                                                   |
| <b>10.3</b> | <b>Possibility of hazardous reactions</b> | Oxidizing agents may cause exothermic reactions.                         |
| <b>10.4</b> | <b>Conditions to avoid</b>                | Avoid extreme temperatures. Avoid humidity.                              |
| <b>10.5</b> | <b>Incompatible materials</b>             | Substances to avoid: strong acids, strong bases, strong oxidizing agents |
| <b>10.6</b> | <b>Hazardous decomposition products:</b>  | No hazardous reactions if stored and handled as prescribed/indicated.    |

### SECTION 11: Toxicological information

|             |                                                |                                                        |
|-------------|------------------------------------------------|--------------------------------------------------------|
| <b>11.1</b> | <b>Information on the product as supplied:</b> |                                                        |
|             | Acute oral toxicity:                           | LD50/oral/rat > 5000 mg/kg.                            |
|             | Acute dermal toxicity:                         | LD50/dermal/rat > 5000 mg/kg.                          |
|             | Acute inhalation toxicity:                     | The product is not expected to be toxic by inhalation. |
|             | Skin corrosion/irritation:                     | Not irritating.                                        |

## VIVAFLOC CP610

Revision: 02

Revision date 19/02/2022

|                                    |                                                                                                                                                                                                                                    |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serious eye damage/eye irritation: | Testing conducted according to the Draize technique showed the material produces no corneal or iridial effects and only slight transitory conjunctival effects similar to those which all granular materials have on conjunctivae. |
| Respiratory/skin sensitization:    | The results of testing on guinea pigs showed this material to be non-sensitizing.                                                                                                                                                  |
| Mutagenicity:                      | Not mutagenic.                                                                                                                                                                                                                     |
| Carcinogenicity:                   | Not carcinogenic.                                                                                                                                                                                                                  |
| Reproductive toxicity:             | Not toxic for reproduction.                                                                                                                                                                                                        |
| STOT – Single exposure:            | No known effects                                                                                                                                                                                                                   |
| STOT – Repeated exposure:          | No known effects                                                                                                                                                                                                                   |
| Aspiration hazard:                 | No hazards resulting from the material as supplied.                                                                                                                                                                                |

### SECTION 12: Ecological information

|                                         |                                                                                                                                                                                                  |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>12.1</b>                             | <b>Toxicity</b>                                                                                                                                                                                  |
| Toxicity to fish:                       | LC50/Danio rerio/96hours= 10- 100 mg/L (OECD 203)                                                                                                                                                |
| Aquatic invertebrates:                  | EC50/Daphnia magna/48 hours> 50 mg/L. (OECD 202)                                                                                                                                                 |
| Acute toxicity to algae:                | Algal inhibition tests are not appropriate. The flocculation characteristics of the product interfere directly in the test medium preventing homogenous distribution which invalidates the test. |
| Chronic toxicity to fish:               | No data available                                                                                                                                                                                |
| Chronic toxicity to invertebrates:      | No data available                                                                                                                                                                                |
| Toxicity to microorganisms:             | No data available                                                                                                                                                                                |
| Effects on terrestrial organisms:       | No data available                                                                                                                                                                                |
| Sediment toxicity:                      | No data available                                                                                                                                                                                |
| <b>12.2</b>                             | <b>Persistence and degradability</b>                                                                                                                                                             |
| Degradation:                            | Readily biodegradable.                                                                                                                                                                           |
| Hydrolysis:                             | At natural pHs (>6) the polymer degrades due to hydrolysis to more than 70% in 28 days. The hydrolysis products are not harmful to aquatic organisms.                                            |
| Photolysis:                             | No data available.                                                                                                                                                                               |
| <b>12.3</b>                             | <b>Bioaccumulation potential</b>                                                                                                                                                                 |
| Information on the product as supplied: | The product is not expected to bioaccumulate.                                                                                                                                                    |

## VIVAFLOC CP610

Revision: 02

Revision date 19/02/2022

|      |                                           |                                                                                                                                                          |
|------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | Partition co-efficient (Log Pow):         | <0                                                                                                                                                       |
|      | Bioconcentration factor (BCF):            | No data available.                                                                                                                                       |
| 12.4 | <b>Mobility in soil</b>                   | No data available.                                                                                                                                       |
| 12.5 | <b>Results of PBT and vPvB assessment</b> | PBT assessment: Not PBT according to the criteria of Annex XIII of REACH.<br>vPvB assessment: Not vPvB according to the criteria of Annex XIII of REACH. |
| 12.6 | <b>Other adverse effects</b>              | None know                                                                                                                                                |

### SECTION 13: Disposal considerations

|      |                                      |                                                                                                                                                                                                                                                          |
|------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13.1 | <b>Waste treatment methods</b>       |                                                                                                                                                                                                                                                          |
|      | Waste from residues/unused products: | Dispose in accordance with local and national regulations. Can be landfilled or incinerated, when in compliance with local regulations.                                                                                                                  |
|      | Contaminated packaging:              | Rinse empty containers with water and use the rinse-water to prepare the working solution. If recycling is not practicable, dispose of in compliance with local regulations. Can be landfilled or incinerated, when in compliance with local regulation. |
|      | Recycling:                           | In accordance with local and national regulations.                                                                                                                                                                                                       |

### SECTION 14: Transport information

|      |                       |                                                                |
|------|-----------------------|----------------------------------------------------------------|
| 14.1 | <b>Land transport</b> |                                                                |
|      | ADR, RID              | Not classified as a dangerous good under transport regulations |
| 14.2 | <b>Sea transport</b>  |                                                                |
|      | IMDG                  | Not classified as a dangerous good under transport regulations |
| 14.3 | <b>Air transport</b>  |                                                                |
|      | IATA/ICAO             | Not classified as a dangerous good under transport regulations |

### SECTION 15: Regulatory information

|      |                                                                                                                                                                                                                                                                                          |  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 15.1 | <b>Safety, health and environmental regulations/legislation specific for the substance or mixture</b>                                                                                                                                                                                    |  |
|      | All components of this product have been registered or pre-registered with the European Chemicals Agency or are exempt from registration.                                                                                                                                                |  |
| 15.2 | <b>Chemical Safety Assessment</b>                                                                                                                                                                                                                                                        |  |
|      | A Chemical Safety Assessment for this product has been carried out by the person responsible for producing this Safety Data sheet. All relevant information used to conduct this assessment are included in this Safety Data Sheet as well any as any resulting Risk Reduction Measures. |  |

| SECTION 16: Other information |                                                                            |                                                                                                                                                                                                                              |
|-------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | Key or legend to abbreviations and acronyms used in the safety data sheet: | PBT = persistent, bioaccumulative and toxic<br>STOT = Specific target organ toxicity<br>vPvB = very persistent and very bioaccumulative                                                                                      |
|                               | Abbreviations                                                              | Eye Irrit. 2 = Serious eye damage/eye irritation, Hazard Category 2<br>Skin Irrit.2 = Skin corrosion/ irritation, Hazard Category 2<br>Aquatic Chronic 3 = Hazardous to the aquatic environment – Chronic Hazard, Category 3 |
|                               | Hazard statements                                                          | H319 – Causes serious eye irritation<br>H315 – Causes skin irritation<br>H412 – Harmful to aquatic life with long lasting effects                                                                                            |
|                               | This SDS was prepared in accordance with the following:                    | Regulation (EC) N°1907/2006, as amended<br>Regulation (EC) N°1272/2008, as amended                                                                                                                                           |

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. The data do not describe the product's properties (product specification). Neither should any agreed property nor the suitability of the product for any specific purpose be deduced from the data contained in the safety data sheet. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

END OF DATA SHEET.